Summary

- A differential equation is an equation with a function and one or more of its derivatives. (i.e) an equation with the function $y = f(x)$ and its derivatives $\frac{dy}{dx}, \frac{d^2y}{dx^2} \ldots$ is called differential equation.
- Order of the highest order derivative present in the differential equation is the order of the differential equation.
- Degree is the highest power of the highest order derivative in the differential equation, after the equation has been cleared from fractions and the radicals as for as the derivatives are concerned.
- A function which satisfies the given differential equation is called its solution. The solution which contains as many arbitrary constants as the order of the differential equation is called a general solution and the solution free from arbitrary constants is called particular solution.
- To form a differential equation from a given function we differentiate the function successively as many times as the number of arbitrary constants in the given function and then eliminate the arbitrary constants.
- In an equation it is possible to collect all the terns of x and dx on one side and all the terms of y and dy on the other side, then the variables are said to be separable. Thus the general form of such an equation is $f(x)dx = g(y)dy$ (or) $f(x)dx + g(y)dy = 0$ By direct integration, we get the solution.
- A differential equation which can be expressed in the form $\frac{dy}{dx} = f(x, y)$ or $\frac{dx}{dy} = g(x, y)$ where $f(x, y)$ and $g(x, y)$ are homogeneous function of degree zero is called a homogeneous differential equation.
- A differential equation of the form $\frac{dy}{dx} + Py = Q$ where P and Q are constants or functions of x only is called a first order linear differential equation.
- A general second order linear differential equation with constant coefficients is of the form $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$

GLOSSARY (கலைச்சொற்கள்)

Abscissa	கிடை அச்சுத் தொலைவு
Arbitrary constant	மாறத்தக்க மாறிலி
Auxiliary equation	துணைச் சமன்பாடு
Complementary function	நிரப்புச் சார்பு
Constant	மாறிலி
Degree	படி
Differential equation	வகைக்கெழுச் சமன்பாடுகள்
Fixed constant	நிலையான மாறிலி

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